Chapter 7

Conclusions and Future Work

7.1 Conclusions

As discussed in Chapter 1, it is the emphasis on multiple principles in designing the image/video information retrieval techniques that constitutes the spirit of this dissertation. Through the dissertation, we practice this design strategy in investigating three related fields: feature computation, indexing of feature space and interactive search and emphasis the mutual impact of these fields. Among the three fields, we identify that the difficulty in indexing the high dimensional feature space and the large amount of image/video items are two dominant factors that motivate the techniques presented in this dissertation. Essentially, it is the task of handling large amount of items giving rise to the need of developing index structures to manage them.

For the feature computation, driven by the difficulty of indexing high dimensional space, we not only consider the effectiveness of the descriptor in characterizing the intended feature, also the feasibility for the descriptor to be organized by the database management system. Under this spirit, in Chapter 3, a compact descriptor for visual browsing of texture database is presented. It demonstrates its effectiveness in browsing large amount of texture patterns and enhancing the similarity search. Meanwhile, the low dimensionality of the descriptor assures that the descriptor can be ingested and managed by DBMS efficiently.

For the high dimensional descriptors used in similarity search, we address the issue of dimensionality reduction, which is also demanded by the challenge of high dimensional indexing. Different from developing general dimensionality reduction techniques, we empha-

sis the impact of user's behavior and interest in similarity search, which leads us to focus on the preservation of local topology of the feature space. Such consideration brings a direct computational advantage for the dimensionality reduction technique presented in Chapter 4. Also in Chapter 4, two performance measures are proposed to evaluate dimensionality reduction techniques. The essential need for such performance measures is to capture human's perceptual preference in measuring the preservation of the local topology.

In Chapter 5, we address the issue of indexing high dimensional space directly. Instead of working on a index structure for general purposes, we focus on the specialties of the feature space consisting of image/video descriptors. The way to access the descriptor in image search techniques, especially the need for interactive learning, suggests the importance of maintaining the index structure to be independent of similarity measures; The investigation on the distribution of the descriptors in high dimensional space demonstrates the variety of the distribution of image descriptors. Based on these considerations and the recent observation of using tree-based index structures for similarity search, we propose an approximation based index structure for similarity search in high dimensional space for large image databases. By choosing the approximation based index structure, various similarity measures can be supported. By efficiently estimating the marginal distribution of the feature space, the index structure can adapt to the distribution of the feature space adaptively. The evaluation on a large image database demonstrates the effectiveness of the proposed indexing scheme.

In considering the image search technique, we acknowledge the importance of the interactive learning in retrieving valuable information from image objects. Keeping in mind the difficulty of indexing high dimensional space, such acknowledgement naturally leads to the need of an index scheme that can support the interactive learning efficiently. Specifically, we focus on a popular interactive learning scenario, relevance feedback, in which iterative nearest neighbor search over varying distance measure is required. For such task, an novel indexing scheme is proposed in Chapter 6. The main idea is to bound the impact of changing distance measure on the nearest neighbor search. The introduced bounds provide the ability to reduce the searching space, which in turn reduces the time to response the interactive learn-

ing. The effectiveness of the proposed indexing scheme is confirmed by the evaluation over a database consisting of around 700,000 color images collected from internet.

The presented techniques aims at enriching the means to retrieve image/video information. The achievement of this goal is verified by the thorough evaluations provided in the dissertation. We attribute such accomplishment to the use of multiple principles in designing the specific techniques.

7.2 Future Work

Image/video information retrieval, as both a research topic and a industry practice, has been of interest for almost a decade. Nevertheless, it is still a long way to go to really understand the market value of the information to be retrieved and how to retrieve it. In the last decade, a lot of effort has been dedicated to explore various possibilities in these two areas, as introduced in Chapter 1. In this dissertation, we focus on the interaction between the visual analysis and database indexing. Along this direction, there are many other open areas to be investigated. The following are two examples.

Scalable learning: In utilizing image/video information, it has been generally recognized that the low level features can only provide a rigid accessibility for information retrieval. For this reason, extensive learning algorithms have been investigated to adapt the feature space, similarity measures to user's expectation. One example is the relevance feedback as introduced in this dissertation. More complicated examples of learning algorithms include using classical Bayesian learning, Support Vector Machine, Multiple-Instance learning and so on. Since these learning algorithms intend to be applied on large image/video databases, it is critical to investigate how the learning performance scales with the size of the database, the dimensionality of the feature space. Moreover, since a learning algorithm needs to access the feature space through the database management system, it is also important to consider the cost caused by indexing the feature space in evaluating the complexity and the effectiveness of the learning algorithms. But the current status in developing learning algorithms for image/video information retrieval usually ignores these issues, which probably

partially explains why we haven not seen any learning algorithm “working in reality” for image/video information retrieval.

Scalable computation and indexing: As more and more devices, such PDA products, available for wireless communication, the multimedia information will be accessed in various ways. For example, Virage Inc. recently partnered with PacketVideo to deliver searchable multimedia over wireless telephones and PDAs. For the “digital classroom” project currently developed in University of California, Santa Barbara, students and instructors may access the teaching materials though a campus computer having LAN connection or a wireless laptop. The diversity of means to support the communication between the users and the multimedia database poses profound impacts on the computation of the feature space and the design of index structures. Just as MPEG-2 and MPEG-4 providing scalable coding streams to adapt to the various bandwidth, it is also desirable to provide a scalable feature computation and a scalable index scheme for different utilizations of multimedia information. For example, a user that has a LAN connection can perform a very detailed search for which the computation uses the descriptors of full dimensions. But for a user using wireless phone, he or she may demand a quick response to a query due to the limitations of power supply and bandwidth. For such case, it requires to use only some important dimensions of the descriptor to compute and index the retrievals so that the quality of service (QoS) can be optimally maintained. The research issues for such occasions need to be identified and addressed.

These open areas provide us the room for our future research on image/video information retrieval.